

CURVATURE AND RADIUS OF CURVATURE

Every curve bends in general and the bending may be different at different points of curve. For example, in case of parabola the bending is different at its different points, being maximum at its vertex. Actually, the curvature is a numerical measure of bending of the curve.

Let P be any point of a curve and s be the length of the arc from a fixed point A of the curve to the point P . Let the tangent at P be inclined at an angle ψ with the axis of x and arc $AP = s$. (See the figure). The direction of the curve at a point is determined by the slope of the tangent at that point. Let Q be another point on the curve close to P so that arc $AQ = s + \delta s$, and the tangent at Q make an angle $\psi + \delta\psi$ with the axis of x . The measure of the rapidity with which the curve is turning (or bending) at the point is the rate of change of ψ with respect to s . This rate of change is called the curvature at the particular point. Thus,

$$\text{The curvature at } P = \lim_{Q \rightarrow P} \frac{\delta\psi}{\delta s} = \lim_{\delta s \rightarrow 0} \frac{\delta\psi}{\delta s} = \frac{d\psi}{ds}.$$

For example, consider a circle of radius a and centre on the y -axis at $(0, a)$, then we have $s = a\psi$ and therefore the curvature $= \frac{d\psi}{ds} = \frac{1}{a}$.

This shows that the curvature of a circle is same at every point of its circumference and is reciprocal of the radius of the circle. Idea is extended to any curve and this is the reason that the reciprocal of the curvature of a curve at a point is defined as its **radius of curvature** at that point and is denoted by ρ . Therefore, we have

$$\frac{1}{\rho} = \frac{d\psi}{ds} \quad \text{or} \quad \rho = \frac{ds}{d\psi}$$

Obviously, the radius of curvature of circle at any point is same as its radius.

RADIUS OF CURVATURE (in Cartesian Coordinates):

Since $\tan \psi = \frac{dy}{dx}$. Differentiating on both sides with respect to s , gives

$$\sec^2 \psi \frac{d\psi}{ds} = \frac{d}{ds} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2} \cdot \frac{dx}{ds}$$

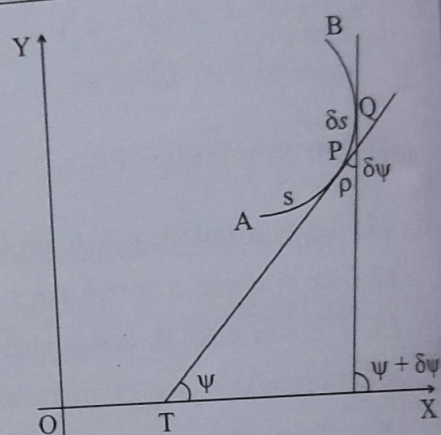
$$\text{or} \quad \frac{ds}{d\psi} = \frac{\sec^2 \psi}{\frac{d^2y}{dx^2}} \cdot \frac{dx}{ds} = \frac{(1 + \tan^2 \psi)}{\frac{d^2y}{dx^2}} \cdot \frac{1}{\sqrt{1 + \left(\frac{dy}{dx} \right)^2}} \quad \text{since} \quad \frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx} \right)^2}.$$

$$\text{or} \quad \rho = \frac{(1 + y'^2)^{3/2}}{y''} \quad \text{where} \quad y' = \frac{dy}{dx} \quad \text{and} \quad y'' = \frac{d^2y}{dx^2} \neq 0$$

In different notation we write $\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$, $y_2 \neq 0$.

In case, the tangent to the curve at the point under consideration is parallel to y -axis then $\frac{dy}{dx}$ will be some

what infinite, hence $\frac{dx}{dy} = 0$ then we consider $\frac{dx}{dy} = \cot \psi$. Then $\rho = \frac{(1 + x_1^2)^{3/2}}{x_2}$ where $x_1 = \frac{dx}{dy}$, $x_2 = \frac{x^2 x}{dy^2} \neq 0$.



RADIUS OF CURVATURE FOR PARAMETRIC CURVES

Let the equation of the curve be $x = f(t)$ and $y = g(t)$ where t is the parameter,

then $\frac{dy}{dx} = \frac{dy}{dt} \bigg/ \frac{dx}{dt} = \frac{y'}{x'}$ where the dashe represents the derivative w.r.t. t .

Further, $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{y'}{x'} \right) = \frac{d}{dt} \left(\frac{y'}{x'} \right) \frac{dt}{dx} = \frac{x' y'' - y' x''}{x'^2 \cdot \frac{dx}{dt}}$

$$\therefore \rho = \frac{\left(1 + \frac{y'^2}{x'^2} \right)^{\frac{3}{2}}}{\frac{x' y'' - y' x''}{x'^3}} = \frac{(x'^2 + y'^2)^{\frac{3}{2}}}{x' y'' - y' x''} \quad \text{or} \quad \rho = \frac{(x_1^2 + y_1^2)^{3/2}}{x_1 y_2 - x_2 y_1}.$$

RADIUS OF CURVATURE AT THE ORIGIN BY NEWTON'S METHOD

This is a very fundamental method but has limited use because it is applicable *only when the curve passes through the origin and has x -axis or y -axis as the tangent there.*

Let us first consider the case where the curve passes through the origin and x -axis is the tangent to the

curve at the origin. Then, $y_1 = 0$ at the origin and hence $\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{1}{y_2}$.

Now let us consider $\lim_{x \rightarrow 0} \frac{x^2}{2y}$ which is of the form $\frac{0}{0}$.

Applying L'Hospital's rule, we get $\lim_{x \rightarrow 0} \frac{x^2}{2y} = \lim_{x \rightarrow 0} \frac{2x}{2y_1}$ which is again of $\frac{0}{0}$ form

Again applying L'Hospital's rule, we get $\lim_{x \rightarrow 0} \frac{x^2}{2y} = \lim_{x \rightarrow 0} \frac{2}{2y_2} = \rho$ {where $y_2 \neq 0$ at $(0, 0)$ }

Therefore, we have $\rho = \lim_{x \rightarrow 0} \frac{x^2}{2y}$, when x -axis is tangent to curve at the origin.

Similarly, in the case when curve passes through the origin and y -axis is the tangent to the curve at the

origin, we have $\rho = \lim_{y \rightarrow 0} \frac{y^2}{2x}$.

EXAMPLE 2.40. Find the radius of curvature at the origin for the following curves

(i) $x^2 = 4ay$

(ii) $y^2(a^2 - x^2) = a^3x$.

SOLUTION: (i) The parabola $x^2 = 4ay$ passes through the origin and tangent at the origin, is

$y = 0$, the axis of x . Therefore, by Newton's method, $\rho = \lim_{x \rightarrow 0} \frac{x^2}{2y} = \lim_{x \rightarrow 0} \frac{4ay}{2y} = 2a$

(ii) The given curve $y^2(a^2 - x^2) = a^3x$ passes through the origin and by equating to 0 the lowest degree term we get the tangent at the origin as $x = 0$, the axis of y . Therefore, by Newton's method,

$$\rho = \lim_{y \rightarrow 0} \frac{y^2}{2x} = \lim_{y \rightarrow 0} \frac{a^3 x}{2(a^2 - x^2) \cdot x} = \lim_{x \rightarrow 0} \frac{a^3}{2a^2 - 2x^2} = \frac{a}{2}, \quad \text{Ans.}$$

EXAMPLE 2.41. If ρ is the radius of curvature at any point P on the parabola $y^2 = 4ax$ and S is its focus then show that ρ^2 varies as SP^3 . Also, show that the radius of curvature at the vertex is equal to the length of the semi-latus rectum. [GGSIPU 2012]

SOLUTION: Differentiating $y^2 = 4ax$ we get $2y y_1 = 4a$ or $y_1 = \frac{2a}{y}$.

and $y_2 = \frac{-2a}{y^2} \cdot y_1 = \frac{-2a}{y^2} \cdot \frac{2a}{y} = \frac{-4a^2}{y^3}$

$$\therefore \rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{\left(1 + 4\frac{a^2}{y^2}\right)^{3/2}}{\frac{-4a^2}{y^3}} = \frac{(y^2 + 4a^2)^{3/2}}{-4a^2}$$

$$= \frac{(4ax + 4a^2)^{3/2}}{4a^2} = \frac{(4a)^{3/2}}{4a^2} (x + a)^{3/2} = \frac{2}{\sqrt{a}} (x + a)^{3/2} \text{ (in absolute value).}$$

Since the focus S is at $(a, 0)$ and $P(x, y)$ is on the parabola, hence

$$SP = \sqrt{(x-a)^2 + (y-0)^2} = \sqrt{(x-a)^2 + 4ax} = x + a$$

$$\therefore \rho = \frac{2}{\sqrt{a}} SP^{3/2} \quad \text{or} \quad \rho^2 = \frac{4}{a} SP^3 \Rightarrow \rho^2 \text{ varies as } SP^3.$$

Next, to obtain ρ at the vertex $(0, 0)$, we find that $\frac{dy}{dx}$ is infinite at the vertex since tangent there, is parallel to y -axis. Therefore, using the Newton's formula

$$\rho = \lim_{y \rightarrow 0} \frac{y^2}{2x} = \lim_{y \rightarrow 0} \frac{4ax}{2x} = 2a = \text{length of semi-latus rectum.}$$

Hence Proved

EXAMPLE 2.42. Find the radius of curvature at any point ' θ ' of the cycloid

$$x = a(\theta + \sin \theta), \quad y = a(1 - \cos \theta).$$

[GGSIPU 2009; 2013; 2014]

SOLUTION: $x' = \frac{dx}{d\theta} = a(1 + \cos \theta), \quad y' = \frac{dy}{d\theta} = a \sin \theta$

$$x'' = \frac{d^2x}{d\theta^2} = -a \sin \theta, \quad y'' = \frac{d^2y}{d\theta^2} = a \cos \theta$$

For parametric equation of the curve, we have

$$\begin{aligned} \rho &= \frac{(x'^2 + y'^2)^{3/2}}{x' y'' - y' x''} = \frac{\{a^2(1 + \cos \theta)^2 + a^2 \sin^2 \theta\}^{3/2}}{a^2(1 + \cos \theta) \cos \theta + a^2 \sin^2 \theta} \\ &= \frac{a(1 + \cos^2 \theta + 2 \cos \theta + \sin^2 \theta)^{3/2}}{\cos \theta + \cos^2 \theta + \sin^2 \theta} = \frac{a(2 + 2 \cos \theta)^{3/2}}{1 + \cos \theta} \quad \therefore \rho = 4a \cos \left(\frac{\theta}{2} \right) \text{ Ans.} \end{aligned}$$

EXAMPLE 2.43. (a) Show that the radius of curvature at any point (x, y) of the rectangular hyperbola

$$xy = c^2, \text{ is given by } \rho = \frac{(x^2 + y^2)^{3/2}}{2c^2}.$$

(b) If ρ_1 and ρ_2 are the radii of curvature at the extremities of a focal chord of parabola, show that $\rho_1^{-2/3} + \rho_2^{-2/3}$ is a constant. [GGSIPU 2011]

SOLUTION: (a) The equation of the curve can be written as $y = c^2/x$

$$\therefore y_1 = \frac{-c^2}{x^2} \quad \text{and} \quad y_2 = \frac{2c^2}{x^3}$$

$$\begin{aligned} \therefore \rho &= \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\left(1 + \frac{c^4}{x^4}\right)^{3/2}}{\frac{2c^2}{x^3}} = \frac{(x^4 + c^4)^{3/2} x^3}{x^6 \cdot 2c^2} = \frac{(x^4 + c^4)^{3/2}}{2c^2 x^3} \\ &= \frac{(x^4 + x^2 y^2)^{3/2}}{2c^2 x^3} = \frac{(x^2 + y^2)^{3/2}}{2c^2} \quad (\text{using } c^2 = xy) \end{aligned}$$

Hence Proved.

(b) Let the parabola be $y^2 = 4ax$, any point on it is $(at^2, 2at)$.

We know that if $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ are the extremities of a focal chord of the parabola, then $t_1 t_2 = -1$.

We have $x = at^2$, $y = 2at$ hence $x' = 2at$, $y' = 2a$ and $x'' = 2a$, $y'' = 0$ therefore at the point $(at^2, 2at)$

$$\rho = \frac{(x'^2 + y'^2)^{3/2}}{x' y'' - y' x''} = \frac{(4a^2 t^2 + 4a^2)^{3/2}}{0 - 4a^2} = 2a(1+t^2)^{3/2} \quad (\text{not taking } - \text{ sign}).$$

$$\text{or} \quad \rho^{-2/3} = (2a)^{-2/3} (1+t^2)^{-1}$$

$$\begin{aligned} \text{Therefore} \quad \rho_1^{-2/3} + \rho_2^{-2/3} &= (2a)^{-2/3} \left(\frac{1}{1+t_1^2} + \frac{1}{1+t_2^2} \right) = (2a)^{-2/3} \frac{(2+t_1^2+t_2^2)}{(1+t_1^2+t_2^2+t_1^2 t_2^2)} \\ &= (2a)^{-2/3} \quad \text{since } t_1 t_2 = -1 = \text{Constant.} \quad \text{Ans.} \end{aligned}$$

EXAMPLE 2.44. If ρ and ρ' are the radii of curvature at the extremities of two conjugate diameters

of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, prove that $(\rho^{2/3} + \rho'^{2/3})(ab)^{2/3} = a^2 + b^2$.

[GGSIPU 2003, 2018]

SOLUTION: Let CP and CD be two conjugate semi diameters of the given ellipse. Parametric equation of the ellipse is $x = a \cos \theta$, $y = b \sin \theta$.

Let the coordinates of the point P be $(a \cos \theta, b \sin \theta)$ then those of the point D are obtained by replacing ' θ ' by $\frac{\pi}{2} + \theta$, so as to get $D = \{a \cos(\theta + \pi/2), b \sin(\theta + \pi/2)\}$. (See the adjoining figure).

To find the radius of curvature at P, the point ' θ ' of the ellipse, we have $x' = -a \sin \theta$, $y' = b \cos \theta$ and $x'' = -a \cos \theta$, $y'' = -b \sin \theta$.

$$\therefore \rho = \frac{(x'^2 + y'^2)^{3/2}}{x' y'' - y' x''} = \frac{(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{3/2}}{(-a \sin \theta)(-b \sin \theta) - (b \cos \theta)(-a \cos \theta)}$$

$$\text{or} \quad \rho = \frac{(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{3/2}}{ab} \quad \dots(1)$$

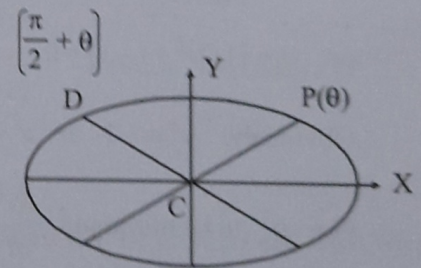
For the radius of curvature ρ' at D, we have to replace θ by $(\theta + \pi/2)$ in (1) and get

$$\rho' = \frac{(a^2 \cos^2 \theta + b^2 \sin^2 \theta)^{3/2}}{ab}$$

$$\therefore \rho^{2/3} + \rho'^{2/3} = \frac{(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{2/3}}{(ab)^{2/3}} + \frac{(a^2 \cos^2 \theta + b^2 \sin^2 \theta)^{2/3}}{(ab)^{2/3}} = \frac{a^2 + b^2}{(ab)^{2/3}}$$

$$\text{or} \quad (ab)^{2/3} (\rho^{2/3} + \rho'^{2/3}) = a^2 + b^2.$$

Hence Proved.



EXAMPLE 2.45. Find the radius of curvature at the origin, for the curve

$$2x^3 - 3x^2y + 4y^3 + y^2 - 3x = 0.$$

SOLUTION: The given curve is passing through the origin and on equating to zero the lowest degree term, we get the equation of the tangent at the origin as $x = 0$, the axis of y .

Therefore, by Newton's method $\rho = \lim_{y \rightarrow 0} \frac{y^2}{2x}$.

Dividing the given equation throughout by x , we get $2x^2 - 3xy + 4y \cdot \frac{y^2}{x} + \frac{y^2}{x} - 3 = 0$.

At the origin, we have $0 + 0 + 0 + 2\rho - 3 = 0$ or $\rho = \frac{3}{2}$. **Ans.**

EXAMPLE 2.46. Find the point on the curve $y = e^x$ at which the curvature is maximum and show that the tangent to the curve at that point forms with the coordinate axes, a triangle whose sides are in the ratio $1 : \sqrt{2} : \sqrt{3}$.

SOLUTION: We have $y = e^x$, hence $y_1 = e^x = y_2 = y$.

$$\therefore \rho = \frac{(1 + y_1^2)^{3/2}}{y_2} \quad \text{or} \quad \rho^2 = \frac{(1 + y_1^2)^3}{y_2^2}.$$

Curvature is maximum when radius of curvature is minimum, i.e., when $\frac{(1 + y_1^2)^3}{y_2^2}$ is minimum.

If we write $y^2 = t$ then $y_1^2 = y_2^2 = t$, hence $\rho^2 = \frac{(1 + t)^3}{t} = t^2 + 3t + 3 + \frac{1}{t}$

Now, $\frac{d\rho^2}{dt} = 0$ gives $2t + 3 - \frac{1}{t^2} = 0$ or $2t^3 + 3t^2 - 1 = 0$

or $(t + 1)(2t^2 + t - 1) = 0$ or $(t + 1)^2(2t - 1) = 0 \Rightarrow t = -1, \frac{1}{2}$ or $y^2 = -1, \frac{1}{2}$

Obviously, $y^2 \neq -1$ as $y^2 \geq 0$, hence $y^2 = \frac{1}{2}$ or $y = \frac{1}{\sqrt{2}}$ since e^x is never negative.

For ρ^2 to be minimum $\frac{d\rho^2}{dt} = 0$ and $\frac{d^2\rho^2}{dt^2} > 0$

$$\frac{d^2\rho^2}{dt^2} = 2 + \frac{2}{t^3} = 2 + \frac{2}{y^6} > 0 \quad \text{hence } \rho^2 \text{ is minimum at } y = \frac{1}{\sqrt{2}}.$$

For $y = \frac{1}{\sqrt{2}}$ we have $x = \log y = -\frac{1}{2} \log 2$.

Thus, the required point is $\left(-\frac{1}{2} \log 2, \frac{1}{\sqrt{2}}\right)$ where the curvature is maximum.

The tangent at the point $\left(-\frac{1}{2} \log 2, \frac{1}{\sqrt{2}}\right)$ has the equation

$$y - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \left(x + \frac{1}{2} \log 2\right) \quad \text{or} \quad \frac{y}{1} - \frac{x}{\sqrt{2}} = \frac{1}{\sqrt{2}} \left(1 + \frac{1}{2} \log 2\right) = k, \text{ say.}$$

Then intercepts on the coordinate axes are $k\sqrt{2}$ and k therefore the tangent is of length

$$= \sqrt{(k\sqrt{2})^2 + k^2} = k\sqrt{3}. \text{ Hence the sides of the triangle are in the ratio } 1 : \sqrt{2} : \sqrt{3}. \text{ Hence Proved.}$$

EXAMPLE 2.47. Show that the radius of curvature at any point of the curve $x^{2/3} + y^{2/3} = a^{2/3}$ is equal to three times the length of the perpendicular from the origin to the tangent at that point. [GGSIPU 2012; 2004; 2010]

SOLUTION: The given curve, in the parametric form, is $x = a \cos^3 \theta, y = a \sin^3 \theta$ hence

$$x' = -3a \cos^2 \theta \sin \theta, y' = 3a \sin^2 \theta \cos \theta$$

$$\text{and } x'' = 6a \cos \theta \sin^2 \theta - 3a \cos^3 \theta, y'' = 6a \sin \theta \cos^2 \theta - 3a \sin^3 \theta.$$

$$\therefore \rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''} = \frac{27a^3 \cos^3 \theta \sin^3 \theta}{-9a^2 \sin^2 \theta \cos^2 \theta} = 3a \sin \theta \cos \theta.$$

The equation of the tangent to the curve at $(a \cos^3 \theta, a \sin^3 \theta)$, is

$$y - a \sin^3 \theta = \frac{-\sin \theta}{\cos \theta} (x - a \cos^3 \theta) \quad \text{or} \quad x \sin \theta + y \cos \theta = a \sin \theta \cos \theta$$

$\therefore$ The length of the perpendicular from the origin upon this perpendicular is $p = a \sin \theta \cos \theta$

Therefore we have $\rho = 3p$.

hence the result.

EXAMPLE 2.48. Find the curvature of $x = 4 \cos t, y = 3 \sin t$. At what point on this ellipse does the curvature have the greatest and the least values? What are magnitudes?

[GGSIPU 2007; 2011]

SOLUTION: The curve is $x = 4 \cos t, y = 3 \sin t$, hence $x' = -4 \sin t, y' = 3 \cos t$,

$$x'' = -4 \cos t, y'' = -3 \sin t.$$

Therefore the radius of curvature ρ at any point 't' on the curve, is given by

$$\rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''} = \frac{(16 \sin^2 t + 9 \cos^2 t)^{3/2}}{-4 \sin t(-3 \sin t) - 3 \cos t(-4 \cos t)} = \frac{1}{12} (9 \cos^2 t + 16 \sin^2 t)^{3/2}$$

$$\text{or } \rho^{2/3} = \frac{1}{12^{2/3}} (9 \cos^2 t + 16 \sin^2 t)$$

Since the curvature is reciprocal of the radius of curvature, the curvature will be greatest and least when the radius of curvature is least and greatest respectively, in other words, when $(12 \rho)^{2/3}$ is least and greatest respectively.

$$\text{For maximum and minimum of } (12 \rho)^{2/3} \text{ we have } \frac{d}{dt} (16 \sin^2 t + 9 \cos^2 t) = 0$$

$$\text{or } 14 \sin t \cos t = 0 \text{ which gives } t = 0 \text{ and } t = \pi/2.$$

Hence $(12 \rho)^{2/3} = 9$ at $t = 0$, i.e. at $(4, 0)$ and is equal to 16 at $t = \pi/2$, i.e. at $(0, 3)$

Thus the curvature is least at $(0, 3)$ and greatest at $(4, 0)$, and least value of curvature is $3/16$ and greatest is $4/9$.

Ans.

EXAMPLE 2.49. Find the radius of curvature of the curve $y^2 = \frac{4a^2(2a-x)}{x}$ at a point where the curve meets the axis of x . [GGSIPU 2007]

SOLUTION: The given curve is $y^2 = \frac{4a^2(2a-x)}{x}$ which meets the axis of x at the point $(2a, 0)$. Since the tangent to the curve at $(2a, 0)$ is parallel to y -axis hence $y_1 = \infty$ at this point. Therefore we consider the other formula $\rho = \frac{(1+x_1^2)^{3/2}}{x_2}$ for finding the radius of curvature at $(2a, 0)$.

The equation of the curve can be written as $x = \frac{8a^3}{y^2 + 4a^2}$

Hence $x_1 = \frac{-16a^3 y}{(y^2 + 4a^2)^2}$ which is 0 at $(2a, 0)$.

$$\begin{aligned} \text{Also } x_2 &= \frac{-16a^3(y^2 + 4a^2)^2 + 16a^3 y \cdot 2(y^2 + 4a^2) \cdot 2y}{(y^2 + 4a^2)^4} \\ &= \frac{-16a^3(y^2 + 4a^2) + 64a^3 y^2}{(y^2 + 4a^2)^3} = \frac{1}{a} \quad \text{at } (2a, 0). \end{aligned}$$

$$\text{Therefore } \rho = \frac{(1+x_1^2)^{3/2}}{x_2} = a \quad \text{at } (2a, 0). \quad \text{Ans.}$$

RADIUS OF CURVATURE (POLAR CURVES)

Let $P(r, \theta)$ be a point on the curve $r = f(\theta)$ and $Q(r + \delta r, \theta + \delta \theta)$ be a neighbouring point on the curve.

Let s denote the length of the arc AP where A is some fixed point on the curve and the arc AQ of length $s + \delta s$ so that arc $PQ = \delta s$. If N is the foot of the perpendicular from P on OQ , then $PN = r \sin \delta \theta$ and $NQ = r + \delta r - r \cos \delta \theta$. (See the figure).

Since $\delta \theta$ is very small we can write $PN = r \delta \theta$ and $NQ = \delta r$.

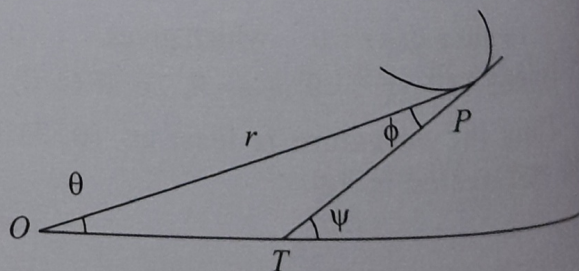
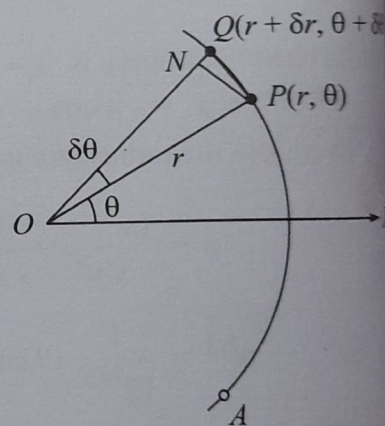
$$\text{Therefore } \delta s = \frac{\text{arc PQ}}{\text{chord PQ}} \cdot \text{chord PQ} = \frac{\text{arc PQ}}{\text{chord PQ}} \sqrt{(r \delta \theta)^2 + (\delta r)^2}$$

$$\Rightarrow \frac{\delta s}{\delta \theta} = \frac{\text{arc PQ}}{\text{chord PQ}} \sqrt{r^2 + \left(\frac{\delta r}{\delta \theta}\right)^2}. \text{ Taking limit as } \delta \theta \rightarrow 0 \text{ we have}$$

$$\frac{ds}{d\theta} = \sqrt{r^2 + r_1^2} \quad \text{where } r_1 = \frac{dr}{d\theta}.$$

Further, let PT be the tangent to the curve at P , making an angle ψ with the initial line OX and ϕ is the angle between OP , the radius vector of P and the tangent there, then, $\psi = \theta + \phi$ (as shown in the figure below).

$$\text{Hence } \frac{d\psi}{d\theta} = 1 + \frac{d\phi}{d\theta}.$$



Differentiating the above relation w.r.t. θ , we get

$$\sec^2 \phi \frac{d\phi}{d\theta} = \frac{r_1 \cdot r_1 - r r_2}{r_1^2} \quad \text{where } r_2 = \frac{d^2 r}{d\theta^2}$$

$$\therefore \frac{d\phi}{d\theta} = \frac{r_1^2 - r r_2}{r_1^2 (1 + \tan^2 \phi)} = \frac{r_1^2 - r r_2}{r_1^2 + r^2} \quad \left(\text{Since } \tan \phi = \frac{r d\theta}{dr} = \frac{r}{r_1} \right)$$

$$\Rightarrow \frac{d\psi}{d\theta} = 1 + \frac{r_1^2 - r r_2}{r_1^2 + r^2} = \frac{2r_1^2 + r^2 - r r_2}{r_1^2 + r^2}$$

Therefore

$$\rho = \frac{ds}{d\psi} = \frac{ds/d\theta}{d\psi/d\theta} = \frac{(r^2 + r_1^2)^{3/2}}{2r_1^2 + r^2 - r r_2}$$

CENTRE OF CURVATURE AND CIRCLE OF CURVATURE

Let APB be a portion of the curve and P(x, y) be any point on it and PT the tangent to the curve at P. If we draw a circle passing through P on the same side of the tangent as the curve, with centre at a point C on the normal to the curve at P so that the radius CP = ρ , the radius of curvature at P, then such a circle is known as *circle of curvature* at P (See figure).

The point C is called the *centre of curvature*. Let $C = (\alpha, \beta)$ then from the figure, we have

$$\alpha = x - \rho \sin \psi \quad \text{and} \quad \beta = y + \rho \cos \psi \quad \text{where } \rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$\text{Since } \tan \psi = \frac{dy}{dx} = y_1, \quad \therefore \sin \psi = \frac{y_1}{\sqrt{1 + y_1^2}} \quad \text{and} \quad \cos \psi = \frac{1}{\sqrt{1 + y_1^2}},$$

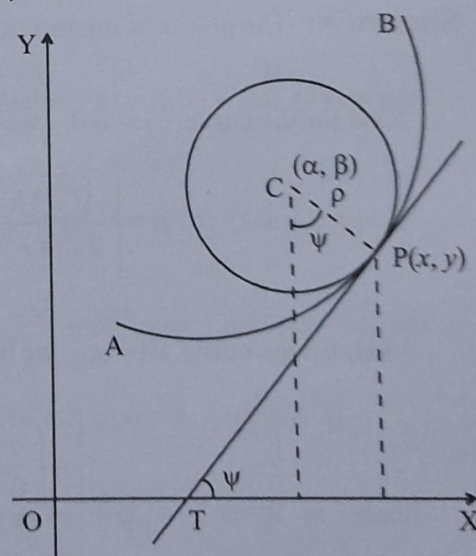
$$\text{therefore } \alpha = x - \frac{(1 + y_1^2)^{3/2}}{y_2} \cdot \frac{y_1}{\sqrt{1 + y_1^2}} = x - \frac{y_1 (1 + y_1^2)}{y_2}$$

$$\text{and } \beta = y + \frac{(1 + y_1^2)^{3/2}}{y_2} \cdot \frac{1}{\sqrt{1 + y_1^2}} = y + \frac{(1 + y_1^2)}{y_2}$$

Thus, the circle of curvature has the equation $(x - \alpha)^2 + (y - \beta)^2 = \rho^2$.

For parametric curves, we have

$$\alpha = x - \frac{y'(x'^2 + y'^2)}{x'y'' - y'x''}, \quad \beta = y + \frac{x'(x'^2 + y'^2)}{x'y'' - y'x''}$$



EXAMPLE 2.50. Find the radius of curvature of the curve $r = a(1 - \cos \theta)$ at the point (r, θ) .

SOLUTION: We have $r = a(1 - \cos \theta)$, hence $r_1 = \frac{dr}{d\theta} = a \sin \theta$ and $r_2 = \frac{d^2 r}{d\theta^2} = a \cos \theta$.

$$\therefore \rho = \frac{(r^2 + r_1^2)^{3/2}}{2r_1^2 + r^2 - r r_2} = \frac{[a^2(1 - \cos \theta)^2 + a^2 \sin^2 \theta]^{3/2}}{2a^2 \sin^2 \theta + a^2(1 - \cos \theta)^2 - a^2(1 - \cos \theta) \cos \theta}$$

$$= \frac{a[1 - 2 \cos \theta + \cos^2 \theta + \sin^2 \theta]^{3/2}}{2 \sin^2 \theta + 1 + 2 \cos^2 \theta - 3 \cos \theta} = \frac{a[2(1 - \cos \theta)]^{3/2}}{3(1 - \cos \theta)} = \frac{4a}{3} \sin \frac{\theta}{2}$$

Also, we can write $\rho = \frac{2a}{3} \sqrt{2(1 - \cos \theta)} = \frac{2}{3} \sqrt{2ar}$ which implies that ρ varies as $\sqrt{r}$.

EXAMPLE 2.51. Show that the radius of curvature at the point (r, θ) of the curve

$$r^n = a^n \cos n\theta, \quad \text{is} \quad \frac{a^n r^{1-n}}{1+n}.$$

SOLUTION: We have $r^n = a^n \cos n\theta$ hence $n r^{n-1} r_1 = -n a^n \sin n\theta$

$$\text{or} \quad r_1 = \frac{-a^n \sin n\theta}{r^n} \quad r = -\frac{-r a^n \sin n\theta}{a^n \cos n\theta} = -r \tan n\theta.$$

$$\text{and} \quad r_2 = -r_1 \tan n\theta - r \sec^2 n\theta \cdot n = r \tan^2 n\theta - n r \sec^2 n\theta.$$

$$\begin{aligned} \therefore \rho &= \frac{(r^2 + r_1^2)^{3/2}}{2r_1^2 + r^2 - r r_2} = \frac{(r^2 + r^2 \tan^2 n\theta)^{3/2}}{2r^2 \tan^2 n\theta + r^2 - r^2 \tan^2 n\theta + n r^2 \sec^2 n\theta} \\ &= \frac{r \sec^3 n\theta}{\tan^2 n\theta + 1 + n \sec^2 n\theta} = \frac{r \sec n\theta}{1+n} = \frac{r^n \sec n\theta}{(1+n) r^{n-1}} = \frac{a^n r^{1-n}}{n+1}. \end{aligned}$$

Hence the res

EXAMPLE 2.52. Show that at the point of intersection of the curves $r = a\theta$ and $r\theta = a$ the curvatures are in the ratio 3 : 1 ($0 < \theta < 2\pi$).

SOLUTION: The points of intersection of the curves $r = a\theta$ and $r\theta = a$, are given by

$$a\theta^2 = a \quad \text{or} \quad \theta = \pm 1.$$

Now for the curve $r = a\theta$ we have $r_1 = a$ and $r_2 = 0$

$$\therefore \text{ at } \theta = \pm 1, \quad \rho = \left[\frac{(r^2 + r_1^2)^{3/2}}{2r_1^2 + r^2 - r r_2} \right]_{\theta = \pm 1} = \left[\frac{(a^2 \theta^2 + a^2)^{3/2}}{2a^2 + a^2 \theta^2 - 0} \right]_{\theta = \pm 1} = \frac{a(2\sqrt{2})}{3} = \rho_1, \text{ say.}$$

Next, for the curve $r\theta = a$, we have $r_1 = \frac{-a}{\theta^2}$ and $r_2 = \frac{2a}{\theta^3}$,

$$\text{hence at } \theta = \pm 1 \quad \rho = \left[\frac{\left(\frac{a^2}{\theta^2} + \frac{a^2}{\theta^4} \right)^{3/2}}{2 \frac{a^2}{\theta^4} + \frac{a^2}{\theta^2} - \frac{2a^2}{\theta^4}} \right]_{\theta = \pm 1} = \left[\frac{a(1 + \theta^2)^{3/2}}{\theta^4} \right]_{\theta = \pm 1} = 2a\sqrt{2} = \rho_2, \text{ say.}$$

$$\therefore \frac{\rho_2}{\rho_1} = \frac{2a\sqrt{2}}{2a\sqrt{2}/3} = \frac{3}{1}. \quad \text{Thus, } \rho_2 : \rho_1 = 3 : 1.$$

Hence Proved